Variable Selection for Wasserstein Robust Hypothesis Testing

Abstract

Variable selection for Wasserstein robust testing admits an exact finite formulation and a maximum-affine dual representation. Dual couplings generate global cuts.

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we follow the convention in [2] to define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{\mathbb{P}_0} \ell(-f(X)) + \mathbb{E}_{\mathbb{P}_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, and $\ell(\cdot)$ is the generating function that is nonnegative-valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^n$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \leq \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with the ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the concatenation of its coordinates whose indices belong to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization problem

$$\min_{S \subseteq [D], |S| \leq d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Finite reformulation and duality

2.1 Fixed-selection reformulation

Fix $S \subseteq [D]$ and write $V(S) = \min_{f \in \mathcal{F}_S} R(f)$. Assume $\rho_0, \rho_1 > 0$ and that ℓ is finite on $\mathbb{R}$. Following the LFD formulation in the related manuscript listed in [2, Lemma 2, Eq. (7)], relabel the pooled observations as $x_1, \ldots, x_{2n}$, with $x_{kn+i} = x_i^{(k)}$. Set

$$c_{ir}^{k,S} = \|x_i^{(k)}[S] - x_r[S]\|_1, \qquad i \in [n],\ r \in [2n],\ k = 0, 1.$$

The referenced lemma gives the following transport problem for randomized testing on the selected coordinates:

$$\begin{aligned} \max_{p_0, p_1, \pi^0, \pi^1 \geq 0} \quad & \sum_{r=1}^{2n} \min\{p_{0r}, p_{1r}\} \\ \text{subject to} \quad & \sum_{r=1}^{2n} \pi_{ir}^k = \frac{1}{n}, \quad i \in [n],\ k = 0, 1, \\ & p_{kr} = \sum_{i=1}^{n} \pi_{ir}^k, \quad r \in [2n],\ k = 0, 1, \\ & \sum_{i=1}^{n} \sum_{r=1}^{2n} c_{ir}^{k,S} \pi_{ir}^k \leq \rho_k, \quad k = 0, 1. \end{aligned} \tag{3}$$

Here π^k_{ir} transports class-k mass from $x^{(k)}_i[S]$ to $x_r[S]$, and p_{kr} is the resulting mass at that target. Let $\mathcal{T}_k(S)$ denote the nonnegative matrices satisfying the row-sum and transport-budget constraints for class k in (3). The column-sum equalities then determine p_k.

For the generating loss in Section 1, write $\ell_0(s) = \ell(-s)$ and $\ell_1(s) = \ell(s)$. At target r, the optimal weighted loss is $\inf_s\{p_{0r}\ell(-s) + p_{1r}\ell(s)\}$. The corresponding extension of the LFD formulation is

$$\begin{aligned} V(S) &= \max_{\pi^0\in\mathcal{T}_0(S),\,\pi^1\in\mathcal{T}_1(S)} \sum_{r=1}^{2n} \inf_{s\in\mathbb{R}}\{p_{0r}\ell(-s) + p_{1r}\ell(s)\} \\ &= \inf_{u\in\mathbb{R}^{2n}} \sum_{k=0}^{1} \max_{\pi^k\in\mathcal{T}_k(S)} \sum_{i=1}^{n}\sum_{r=1}^{2n} \pi^k_{ir}\ell_k(u_r). \end{aligned}$$

The first equality requires a general-loss extension of the lemma; Appendix A proves it by constructing detectors and transport plans in both directions. The second equality follows from finite-dimensional minimax duality. Its expected-loss term follows directly from the column sums: $\sum_r p_{kr}\ell_k(u_r) = \sum_{i,r}\pi^k_{ir}\ell_k(u_r)$.

For fixed u, dualize the class-k transport LP. Assign $a^k_i \in \mathbb{R}$ to its row-sum constraint and $\lambda_k \geq 0$ to its budget. Its Lagrangian is

$$\frac{1}{n}\sum_i a^k_i + \rho_k\lambda_k + \sum_{i,r}\pi^k_{ir}\left[\ell_k(u_r) - a^k_i - \lambda_k c^{k,S}_{ir}\right].$$

The supremum over $\pi^k \geq 0$ is finite precisely when every bracket is nonpositive. LP strong duality, followed by minimization over u, therefore gives

$$\begin{aligned} V(S) = \min_{a,u,\lambda}\quad & \underbrace{\frac{1}{n}\sum_{k=0}^{1}\sum_{i=1}^{n} a^k_i + \sum_{k=0}^{1}\rho_k\lambda_k}_{J(a,\lambda)} \\ \text{subject to}\quad & \ell_k(u_r) \leq a^k_i + \lambda_k c^{k,S}_{ir}, \quad i\in[n],\ r\in[2n],\ k=0,1, \\ & a\in\mathbb{R}^{2\times n}, \quad u\in\mathbb{R}^{2n}, \quad \lambda\in\mathbb{R}^2_+. \end{aligned} \tag{4}$$

Thus (4) has exactly the optimal risk of detectors using S; minimizing it over $|S| \leq d$ solves the problem in Section 1. The proof in Appendix A also defines an optimal detector outside the pooled observations.

2.2 Mixed-integer reformulation

Let $z = \mathbf{1}_S$ and define the coordinate costs

$$\Delta^k_{ir,j} = |x^{(k)}_i[j] - x_r[j]|, \qquad c^{k,S}_{ir} = \sum_{j\in S}\Delta^k_{ir,j}.$$

Following the logical-constraint formulation of Bertsimas et al. [1], introduce coordinate prices w_{kj} that vanish when $z_j = 0$. For fixed $z \in [0,1]^D$, define

$$\begin{aligned} F(z) = \min_{a,u,\lambda,w}\quad & J(a,\lambda) \\ \text{subject to}\quad & \ell_k(u_r) \leq a^k_i + \sum_{j=1}^{D}\Delta^k_{ir,j}w_{kj}, \quad i\in[n],\ r\in[2n],\ k=0,1, \\ & 0 \leq w_{kj} \leq \lambda_k, \quad k=0,1,\ j\in[D], \\ & \rho_0 w_{0j} + \rho_1 w_{1j} \leq z_j J(a,\lambda), \quad j\in[D], \\ & a\in\mathbb{R}^{2\times n}, \quad u\in\mathbb{R}^{2n}, \quad \lambda\in\mathbb{R}^2_+. \end{aligned} \tag{5}$$

Choosing the source itself as target gives $a^k_i \geq 0$. Consequently, when $z_j = 1$, its selection inequality is redundant:

$$\rho_0 w_{0j} + \rho_1 w_{1j} \leq \rho_0\lambda_0 + \rho_1\lambda_1 \leq J(a,\lambda).$$

When $z_j = 0$, it forces $w_{0j} = w_{1j} = 0$. A feasible solution of (4) lifts to (5) through $w_{kj} = \lambda_k z_j$. Conversely, at $z = \mathbf{1}_S$,

$$\ell_k(u_r) \le a_i^k + \sum_{j \in S} \Delta_{ir,j}^k w_{kj} \le a_i^k + \lambda_k c_{ir}^{k,S}.$$

Both directions preserve J, so $F(\mathbf{1}_S) = V(S)$. This proves the exactness of the reformulation for binary selections.

2.3 Dual representation

Let $\Pi_{ir}^k \ge 0$ and $\mu_j \ge 0$ be multipliers of the loss and selection constraints in (5). The dual objective is

$$H(\Pi) = \sum_{r=1}^{2n} \inf_{s \in \mathbb{R}} \sum_{k=0}^{1} \sum_{i=1}^{n} \Pi_{ir}^k \ell_k(s).$$

It is the sum of the optimal weighted losses at the targets, with weights given by the column masses of Π^0 and Π^1. Minimizing the Lagrangian over a, u, λ, w gives

$$\begin{aligned} F(z) = \max_{\Pi, \alpha, \mu} \quad & H(\Pi) \\ \text{subject to} \quad & (\Pi, \alpha, \mu) \in \mathcal{D}, \qquad \alpha + \mu^\top z = 1, \end{aligned} \tag{6}$$

where $(v)_+ = \max\{v, 0\}$ and

$$\mathcal{D} = \left\{ (\Pi, \alpha, \mu) \ge 0 : \begin{array}{ll} \displaystyle\sum_{r=1}^{2n} \Pi_{ir}^k = \alpha/n, & i \in [n],\ k = 0, 1, \\ \displaystyle\sum_{j=1}^{D} \left(\sum_{i,r} \Pi_{ir}^k \Delta_{ir,j}^k - \rho_k \mu_j \right)_+ \le \alpha \rho_k, & k = 0, 1 \end{array} \right\}. \tag{7}$$

The row sums show that Π^k is a transport plan with its source masses scaled by α. Appendix B derives the positive-part constraints by eliminating the multipliers of $w_{kj} \le \lambda_k$.

The set $\mathcal{D}$ is a cone and $H(\tau\Pi) = \tau H(\Pi)$ for $\tau \ge 0$. For any normalized tuple,

$$\max_{\tau \ge 0} \{2\sqrt{\tau H(\Pi)} - \tau\} = H(\Pi).$$

Optimizing this scale removes the normalization and yields

$$\boxed{F(z) = \max_{(\Pi, \alpha, \mu) \in \mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}.} \tag{8}$$

An optimal normalized tuple $(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$ is multiplied by $F(z)$ to obtain an optimizer of (8). Thus its affine slope is $-F(z)\bar{\mu}$. Proposition B.1 proves this representation and the resulting subgradient formula.

3 Cutting-plane algorithm

The affine representation allows the outer-approximation strategy of Bertsimas et al. [1, Section 3.1]: solve the continuous problem for one selection, retain its affine lower bound, and use a binary linear program to choose the next selection. Write $\mathrm{OPT} = \min_{|S| \le d} V(S)$ and let $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$. At z^t, take an optimizer (Π^t, α^t, μ^t) of (8). It defines

$$\begin{aligned} & c_t(z) = 2\sqrt{H(\Pi^t)} - \alpha^t - (\mu^t)^\top z, \\ & F(z) \ge c_t(z) = F(z^t) - (\mu^t)^\top (z - z^t), \qquad c_t(z^t) = F(z^t). \end{aligned} \tag{9}$$

Dual feasibility gives the lower bound, and dual optimality gives equality at z^t. Thus this is a supporting affine function, not merely an arbitrary lower bound; no differentiability of F is required. The next

selection and the master lower bound solve

$$
\begin{aligned}
(z^{t+1}, \eta^{t+1}) \in \ & \operatorname*{arg\,min}_{z \in \mathcal{Z}_d,\ \eta \geq 0} \quad \eta \\
& \text{subject to} \quad \eta \geq F(z^s) - (\mu^s)^\top (z - z^s), \qquad s = 1, \ldots, t.
\end{aligned}
\tag{10}
$$

With $U_t = \min_{s \leq t} F(z^s)$, all retained cuts lie below F, so

$$
\eta^{t+1} \leq \mathrm{OPT} \leq U_t.
$$

Stop when $U_t - \eta^{t+1} \leq \varepsilon$ and return the best evaluated selection. Appendix A recovers its detector from a solution of the finite convex problem.

Algorithm 1 Exact cutting-plane method for variable selection

Require: Problem data, budget d, tolerance $\varepsilon > 0$, and $z^1 \in \mathcal{Z}_d$.
Ensure: A subset S^*, its optimal detector f^*, and a risk interval of width at most ε.

$t \leftarrow 1, U \leftarrow +\infty$.
while true **do**
  Solve (8) at z^t and take an optimizer (Π^t, α^t, μ^t).
  if $F(z^t) < U$ **then**
    $(U, z^*) \leftarrow (F(z^t), z^t)$.
  end if
  Retain the cut (9) with index t.
  Solve (10) to obtain (z^{t+1}, η^{t+1}).
  if $U - \eta^{t+1} \leq \varepsilon$ **then**
    $S^* \leftarrow \{j : z^*_j = 1\}$.
    Solve (4) for S^* and construct g^* by (14).
    Set $f^*(x) = g^*(x[S^*])$ and **return** $(S^*, f^*, [\eta^{t+1}, U])$.
  end if
  $t \leftarrow t + 1$.
end while

A Finite reformulation

Proposition A.1 (Finite reformulation)**.** *For every $S \subseteq [D]$, the minimum in* (4) *is attained and*

$$
\begin{aligned}
V(S) &= \min_{u \in \mathbb{R}^{2n}} \sum_{k=0}^{1} \max_{\pi^k \in \mathcal{T}_k(S)} \sum_{i=1}^{n} \sum_{r=1}^{2n} \pi^k_{ir} \ell_k(u_r) \\
&= \max_{\substack{\pi^0 \in \mathcal{T}_0(S) \\ \pi^1 \in \mathcal{T}_1(S)}} \sum_{r=1}^{2n} \inf_{s \in \mathbb{R}} \{p_{0r} \ell(-s) + p_{1r} \ell(s)\}.
\end{aligned}
\tag{11}
$$

Here $p_{kr} = \sum_{i=1}^{n} \pi^k_{ir}$. An optimal solution of (4) *also yields an optimal detector $f(x) = g(x[S])$ on $\mathbb{R}^D$.*

Proof. For fixed u and class k, dualize the source-mass constraints of the transport LP with $a^k_i \in \mathbb{R}$ and its budget constraint with $\lambda_k \geq 0$. Its Lagrangian is

$$
\begin{aligned}
L_k &= \sum_{i,r} \pi^k_{ir} \ell_k(u_r) + \sum_i a^k_i \left(\frac{1}{n} - \sum_r \pi^k_{ir} \right) + \lambda_k \left(\rho_k - \sum_{i,r} c^{k,S}_{ir} \pi^k_{ir} \right) \\
&= \frac{1}{n} \sum_i a^k_i + \rho_k \lambda_k + \sum_{i,r} \pi^k_{ir} [\ell_k(u_r) - a^k_i - \lambda_k c^{k,S}_{ir}].
\end{aligned}
\tag{12}
$$

The supremum over $\pi^k \geq 0$ is finite exactly when every bracket is nonpositive. Linear programming duality therefore gives

$$\begin{aligned} \max_{\pi^k \in \mathcal{T}_k(S)} \sum_{i,r} \pi^k_{ir} \ell_k(u_r) = \min_{a^k \in \mathbb{R}^n,\, \lambda_k \geq 0} \quad & \frac{1}{n} \sum_i a^k_i + \rho_k \lambda_k \\ \text{subject to} \quad & \ell_k(u_r) \leq a^k_i + \lambda_k c^{k,S}_{ir}, \quad i \in [n],\ r \in [2n]. \end{aligned} \tag{13}$$

Consequently, the infimum W in (4) equals the infimum of the first line of (11).

For any $f(x) = g(x[S])$, take $u_r = g(x_r[S])$. A plan $\pi^k \in \mathcal{T}_k(S)$ defines a distribution in $\mathcal{P}_k$ by moving mass π^k_{ir} from $x^{(k)}_i$ to the point whose selected coordinates are $x_r[S]$ and whose other coordinates remain $x^{(k)}_i[S^c]$. Its transport cost is $\sum_{i,r} c^{k,S}_{ir} \pi^k_{ir}$, and its expected loss is $\sum_{i,r} \pi^k_{ir} \ell_k(u_r)$. Hence

$$W \leq \sum_{k=0}^{1} \max_{\pi^k \in \mathcal{T}_k(S)} \sum_{i,r} \pi^k_{ir} \ell_k(g(x_r[S])) \leq R(f).$$

Taking the infimum over $f \in \mathcal{F}_S$ proves $W \leq V(S)$.

For the reverse inequality, let (a, u, λ) be feasible in (4). For $y \in \mathbb{R}^{|S|}$, define

$$E_k(y) = \min_{i \in [n]} \{a^k_i + \lambda_k \|x^{(k)}_i[S] - y\|_1\},$$
$$i_k = \min \arg\min_{i \in [n]} \{a^k_i + \lambda_k \|x^{(k)}_i[S] - y\|_1\}.$$

Put $r_k = \|x^{(k)}_{i_k}[S] - y\|_1$ and $b = \|x^{(0)}_{i_0}[S] - x^{(1)}_{i_1}[S]\|_1$. Set

$$g(y) = \begin{cases} (1-\gamma) u_{i_0} + \gamma u_{n+i_1}, & \gamma = \min\{1, r_0/b\}, \quad b > 0, \\ u_{i_0}, & b = 0. \end{cases} \tag{14}$$

For $b > 0$, the triangle inequality gives $\gamma b \leq r_0$ and $(1-\gamma) b \leq r_1$. Convexity and the source–target constraints in (4) yield

$$\ell_0(g(y)) \leq (1-\gamma)\ell_0(u_{i_0}) + \gamma \ell_0(u_{n+i_1}) \leq a^0_{i_0} + \lambda_0 \gamma b \leq E_0(y),$$
$$\ell_1(g(y)) \leq (1-\gamma)\ell_1(u_{i_0}) + \gamma \ell_1(u_{n+i_1}) \leq a^1_{i_1} + \lambda_1 (1-\gamma) b \leq E_1(y).$$

When $b = 0$, the same bounds follow from the two constraints at target i_0. The finite tie rule makes g Borel measurable, and $|g(y)| \leq \max_r |u_r|$. For any $\mathbb{P}_k \in \mathcal{P}_k$, integrating these bounds against a coupling from $\widehat{\mathbb{P}}_k$ to $\mathbb{P}_k$ of cost at most ρ_k gives

$$\mathbb{E}_{\mathbb{P}_k} \ell_k(g(X[S])) \leq \frac{1}{n} \sum_i a^k_i + \rho_k \lambda_k, \qquad R(f) \leq J(a, \lambda), \quad f(x) = g(x[S]).$$

Thus $V(S) \leq W$, proving equality with the finite convex problem.

The product $\mathcal{T}_0(S) \times \mathcal{T}_1(S)$ is compact and convex. The function $\sum_{k,i,r} \pi^k_{ir} \ell_k(u_r)$ is continuous, affine in the plans, and convex in u. Finite-dimensional minimax duality gives

$$\begin{aligned} W &= \inf_u \max_{\pi^0, \pi^1} \sum_{k,i,r} \pi^k_{ir} \ell_k(u_r) = \max_{\pi^0, \pi^1} \inf_u \sum_{k,i,r} \pi^k_{ir} \ell_k(u_r) \\ &= \max_{\pi^0, \pi^1} \sum_r \inf_s \{p_{0r} \ell(-s) + p_{1r} \ell(s)\}, \end{aligned}$$

where each pair of plans ranges over the same product set. This proves the general-loss formulation starting from the transport constraints of (3).

Finally, $(a, u, \lambda) = (\mathbf{1}, 0, 0)$ is feasible with $J = 2$. On the sublevel $J \leq 2$, the diagonal constraints give $0 \leq a^k_i \leq 2n$, while $0 \leq \lambda_k \leq 2/\rho_k$. The remaining constraints bound both $\ell(u_r)$ and $\ell(-u_r)$. Since $\ell(t) \to \infty$ as $t \to \infty$, this sublevel is compact, so the minimum is attained. The recovered detector then attains $V(S)$, and its finite scores attain the first minimum in (11). The last maximum is attained because its objective is upper semicontinuous on the compact transport set. □

B Duality

Proposition B.1 (Exactness and dual representation)**.** (i) *For every $S \subseteq [D]$, $F(\mathbf{1}_S) = V(S)$. For every $z \in [0,1]^D$, the primal minimum and the normalized dual maximum are attained, and*

$$0 < F(z) = \max_{\substack{(\Pi,\alpha,\mu)\in\mathcal{D} \\ \alpha+\mu^\top z=1}} H(\Pi) \le 2. \tag{15}$$

(ii) *The same value is attained in*

$$F(z) = \max_{(\Pi,\alpha,\mu)\in\mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}. \tag{16}$$

If $(\bar\Pi, \bar\alpha, \bar\mu)$ is a normalized optimizer, then an optimizer of (16) *is*

$$(\Pi, \alpha, \mu) = F(z)(\bar\Pi, \bar\alpha, \bar\mu), \qquad H(\Pi) = F(z)^2. \tag{17}$$

(iii) *The function F is convex on $[0,1]^D$. An optimizer (Π^t, α^t, μ^t) of* (16) *at z^t gives*

$$\begin{aligned} c_t(z) &= 2\sqrt{H(\Pi^t)} - \alpha^t - (\mu^t)^\top z \le F(z), \\ c_t(z^t) &= F(z^t), \qquad F(z) \ge F(z^t) - (\mu^t)^\top (z - z^t). \end{aligned} \tag{18}$$

Hence $-\mu^t \in \partial F(z^t)$, with the subgradient inequality understood on $[0,1]^D$.

Proof. (i) Taking the source itself as the target in the loss constraints gives $a_i^k \ge 0$ in both primal formulations. For a feasible (a, u, λ) of (4), set $z = \mathbf{1}_S$ and $w_{kj} = \lambda_k z_j$. Then

$$\ell_k(u_r) \le a_i^k + \sum_{j=1}^{D} \Delta_{ir,j}^k w_{kj}, \qquad \sum_{k=0}^{1} \rho_k w_{kj} = z_j \sum_{k=0}^{1} \rho_k \lambda_k \le z_j J(a, \lambda).$$

Conversely, feasibility in (5) with $z = \mathbf{1}_S$ and $\rho_k > 0$ gives

$$w_{kj} = 0 \quad (j \notin S), \qquad \ell_k(u_r) \le a_i^k + \sum_{j\in S} \Delta_{ir,j}^k w_{kj} \le a_i^k + \lambda_k c_{ir}^{k,S}.$$

Thus both formulations have the same feasible (a, u, λ) after eliminating w, and the same objective. This proves $F(\mathbf{1}_S) = V(S)$.

Assign nonnegative multipliers Π_{ir}^k, β_{kj}, and μ_j to the loss constraints, the inequalities $w_{kj} \le \lambda_k$, and the selection constraints in (5), respectively. Keeping a, u free and $\lambda, w \ge 0$, write $\alpha = 1 - \mu^\top z$. The Lagrangian is

$$\begin{aligned} \mathcal{L} &= J(a,\lambda) + \sum_{k,i,r} \Pi_{ir}^k \left[\ell_k(u_r) - a_i^k - \sum_j \Delta_{ir,j}^k w_{kj} \right] \\ &\quad + \sum_{k,j} \beta_{kj}(w_{kj} - \lambda_k) + \sum_j \mu_j \left[\sum_k \rho_k w_{kj} - z_j J(a,\lambda) \right] \\ &= \sum_{k,i} a_i^k \left(\frac{\alpha}{n} - \sum_r \Pi_{ir}^k \right) + \sum_{r=1}^{2n} \sum_{k=0}^{1} \sum_{i=1}^{n} \Pi_{ir}^k \ell_k(u_r) \\ &\quad + \sum_k \lambda_k \left(\alpha\rho_k - \sum_j \beta_{kj} \right) + \sum_{k,j} w_{kj} \left[\beta_{kj} + \rho_k \mu_j - \sum_{i,r} \Pi_{ir}^k \Delta_{ir,j}^k \right]. \end{aligned} \tag{19}$$

Minimization over a, λ, w is finite if and only if

$$\begin{aligned} &\sum_{r=1}^{2n} \Pi_{ir}^k = \frac{\alpha}{n} \quad (k = 0, 1,\ i \in [n]), \qquad \sum_{j=1}^{D} \beta_{kj} \le \alpha\rho_k \quad (k = 0, 1), \\ &\beta_{kj} \ge \left(\sum_{i=1}^{n} \sum_{r=1}^{2n} \Pi_{ir}^k \Delta_{ir,j}^k - \rho_k \mu_j \right)_+ \quad (k = 0, 1,\ j \in [D]). \end{aligned} \tag{20}$$

The row equalities force $\alpha \geq 0$. Minimization over u separates across the $2n$ targets and yields

$$\inf_u \sum_{r=1}^{2n} \sum_{k=0}^{1} \sum_{i=1}^{n} \Pi_{ir}^k \ell_k(u_r) = \sum_{r=1}^{2n} \inf_{s\in\mathbb{R}} \sum_{k=0}^{1} \sum_{i=1}^{n} \Pi_{ir}^k \ell_k(s) = H(\Pi).$$

Eliminating β from (20) gives

$$\sum_{j=1}^{D} \left(\sum_{i=1}^{n} \sum_{r=1}^{2n} \Pi_{ir}^k \Delta_{ir,j}^k - \rho_k \mu_j \right)_+ \leq \alpha \rho_k, \qquad k = 0, 1.$$

These are precisely the constraints defining $\mathcal{D}$; $\alpha = 1 - \mu^\top z$ supplies the normalization in (15).

To establish strong duality also when some $z_j = 0$, eliminate the corresponding variables w_{kj} and their redundant cap and selection constraints. The reduced problem has the strictly feasible point

$$u = 0, \qquad a_i^k = 2, \qquad \lambda_k = 1, \qquad w_{kj} = z_j/2 \quad (z_j > 0).$$

Indeed, $\ell_k(0) = 1 < 2$, $0 < w_{kj} < 1$, and

$$\sum_{k=0}^{1} \rho_k w_{kj} = \frac{z_j}{2}(\rho_0 + \rho_1) < z_j(4 + \rho_0 + \rho_1) = z_j J(a, \lambda) \quad (z_j > 0).$$

Slater's theorem gives equality of the reduced primal and dual values and attainment of the dual maximum. For each eliminated coordinate, restore the full dual by setting

$$\mu_j = \max_{k=0,1} \frac{1}{\rho_k} \sum_{i=1}^{n} \sum_{r=1}^{2n} \Pi_{ir}^k \Delta_{ir,j}^k, \qquad \beta_{kj} = 0 \quad (z_j = 0).$$

This preserves both the objective and the normalization and satisfies (20). Hence the normalized dual attains $F(z)$ for every $z \in [0, 1]^D$.

The feasible point $a_i^k = 1$, $u = 0$, $\lambda = w = 0$ gives $F(z) \leq 2$. On the sublevel $J \leq 2$, nonnegativity gives $a_i^k \leq 2n$, $\lambda_k \leq 2/\rho_k$, and $w_{kj} \leq 2/\rho_k$. The loss constraints then bound both $\ell(u_r)$ and $\ell(-u_r)$. Convexity, monotonicity, and $\ell(0) = 1$ together with $\lim_{t\to-\infty} \ell(t) = 0$ imply $\ell(t) \to \infty$ as $t \to \infty$. Thus the sublevel is compact, and the primal minimum is attained. If its value were zero, then $a = \lambda = w = 0$, contradicting

$$2 = 2\ell(0) \leq \ell(-u_r) + \ell(u_r) \leq 0.$$

This proves (15).

(ii) The set $\mathcal{D}$ is a cone and $H(\tau\Pi) = \tau H(\Pi) \geq 0$ for $\tau \geq 0$. For $(\Pi, \alpha, \mu) \in \mathcal{D}$, let $b = \alpha + \mu^\top z$. If $b > 0$, dividing the tuple by b and applying part (i) gives $H(\Pi)/b \leq F(z)$. Consequently,

$$2\sqrt{H(\Pi)} - b = \frac{H(\Pi)}{b} - \left(\sqrt{\frac{H(\Pi)}{b}} - \sqrt{b} \right)^2 \leq F(z). \tag{21}$$

If $b = 0$, then $\alpha = 0$ and the row equalities imply $\Pi = 0$, so the affine objective is zero. Conversely, for a normalized optimizer $(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$,

$$H(F(z)\bar{\Pi}) = F(z)^2, \qquad 2\sqrt{H(F(z)\bar{\Pi})} - F(z)(\bar{\alpha} + \bar{\mu}^\top z) = F(z).$$

The affine upper bound is therefore attained, proving (16) and (17).

(iii) The feasible set $\mathcal{D}$ is independent of z, so (16) expresses F as a maximum of affine functions and proves convexity. Feasibility of (Π^t, α^t, μ^t) gives $c_t(z) \leq F(z)$ for every $z \in [0, 1]^D$, while optimality at z^t gives $c_t(z^t) = F(z^t)$. Therefore,

$$F(z) \geq c_t(z) = c_t(z^t) - (\mu^t)^\top (z - z^t) = F(z^t) - (\mu^t)^\top (z - z^t). \qquad \square$$

References

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